

Form, Not Granted

A sparse-autoencoder autopsy finds no native authorization concept at an agent’s commit point: the “authorization direction” is a surface-form artifact

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Abstract

A late-layer “authorization direction” was reported to detect and steer a tool-using agent’s commitment to unauthorized irreversible actions [1], and a follow-up found it tracks the authorization the model *feels* rather than the one the user *granted* [2]. Both were built on a *supervised* difference-of-means direction — a method that returns a direction whether or not the concept is natively represented. We stop using the ruler and point the microscope: we encode the residual at the delete-commit token (layer 59 of Qwen3.6-27B) through a trained TopK sparse autoencoder (40,960 features) and ask, unsupervised, whether a feature distinguishes a *granted* commit from a *felt*-but-not-granted one. Three results. (1) **The supervised direction inverts under a structure-matched control**: applying the same frozen direction, AUROC(granted vs felt) goes from 0.838 (single-turn vs multi-turn conditions) to 0.08 when both conditions share one multi-turn scaffold and only the final user turn varies — a direction measuring authorization would not flip when only the conversation *format* changes. (2) **The SAE’s separating features are surface form**: across runs, every top feature that separates granted from felt is, by its max-activating context, a tool-call scaffold, a JSON tool-schema token, or the file-listing output — never a semantic “user granted this” feature. (3) **A cross-lexical-family test isolates this**: a feature selected on one paraphrase family does *not* transfer to a disjoint family (held-out AUROC 0.50, chance), and the one feature that does transfer (0.99, pooled 0.998) max-activates on tool-call *format boilerplate*, not authorization. The agent’s commit-point representation is dominated by **surface form and plan-coherence**; *authorization-as-granted is not a native monosemantic feature* at this locus. Internal authorization monitors fail not because they are mis-calibrated but because there may be no concept to read. We state the scope plainly (one model, one locus, synthetic scenarios, an SAE explaining ~66% of residual variance) and release every ledger and a recompute-everything evaluation.

1 Introduction

Linear directions in the residual stream detect and steer safety-relevant behavior: refusal lives in a single direction [3], and a late-layer *authorization direction* both separates authorized from unauthorized irreversible-action commits and, under steering, moves the commitment [1]. A follow-up sharpened this to “felt, not granted”: in realistic free generation the same direction tracks the authorization the model *feels* (and is blind to the user’s actual grant) [2].

Every step of that arc used a *supervised* difference-of-means direction $d = \overline{\text{auth}} - \overline{\text{unauth}}$. This method has a known failure mode: it *always* returns a direction that separates the labeled

sets, whether the model represents the concept monosemantically or whether we merely projected a human concept onto whatever variance correlated with the labels [7, 8]. The “felt, not granted” result is itself a hint that the concept is imported. So we ask the question the ruler cannot answer: at the moment an agent commits an irreversible action, does it *represent* “user-granted authorization” in its own ontology — or is the decision substrate something else, onto which “authorization” was projected? We answer it by letting the model’s own features speak, via a sparse autoencoder [4, 5, 6] trained on this exact model.

Contributions: (i) a structure-matched control shows the supervised “authorization direction” is *confounded by conversation structure* — it does not weaken but *inverts* (§3); (ii) an unsupervised SAE autopsy shows the features that separate granted from felt are *surface form*, named by their max-activating contexts (§4); (iii) a cross-lexical-family generalization test isolates lexical/scaffold confounds from semantics and finds no paraphrase-invariant *semantic* feature (§5); (iv) the synthesis — the agent commits on form and plan-coherence, and authorization-as-granted is not a native concept here, which reframes why internal monitors fail (§6). This qualifies, not refutes, [1, 2]: the direction is real and usable in its tested regime, but it reads form, not a represented grant.

2 Setup

The direction and the SAE. Following [1], d is the difference of mean layer-59 residuals at the action-commitment token over explicitly-authorized vs. unauthorized prefilled deletes, frozen. The SAE is a TopK autoencoder (caiovicentino1/qwen36-27b-sae-fullstack, layer 59, $d_{\text{sae}} = 40,960$, $k = 128$; reconstruction relative error 0.58, fraction-of-variance-explained ≈ 0.66). We encode a residual x as $z = \text{TopK}_k(\text{ReLU}((x - b_{\text{dec}})W_{\text{enc}} + b_{\text{enc}}))$.

Conditions. A real Qwen3.6-27B agent (bf16) runs free generation in a sandboxed file workspace and commits `delete_file`; we capture the layer-59 residual at the `<function=` decision token. *granted* = the user explicitly authorizes the deletion; *felt* = the agent finishes a cleanup plan it co-authored under a vague “go ahead” (the realistic over-reach of [2]). $n = 12$ –18 commits per condition; greedy decoding (deterministic; the decisive run reproduced bit-for-bit across two GPU sessions). Ledgers and scripts are public.

3 The supervised “authorization direction” inverts under structure-matching

In a first autopsy the conditions differed in conversation *structure* (granted was single-turn; felt was multi-turn: tidy $\rightarrow$ list $\rightarrow$ flag $\rightarrow$ “go ahead”). A structure-matched control removes this: both conditions share one multi-turn scaffold and differ *only* in the final user turn. We then apply the *same frozen* direction d — no selection, no fitting (Fig. 1):

setting	supervised d AUROC (granted vs felt)
unmatched (structure differs)	0.838
structure-matched	0.08 — it <i>inverts</i>

The same direction gives near-opposite answers depending on prompt format. A direction that measured *granted authorization* would be invariant to changing only the conversation format while holding the authorization meaning fixed. It is not. **The “authorization direction” is**

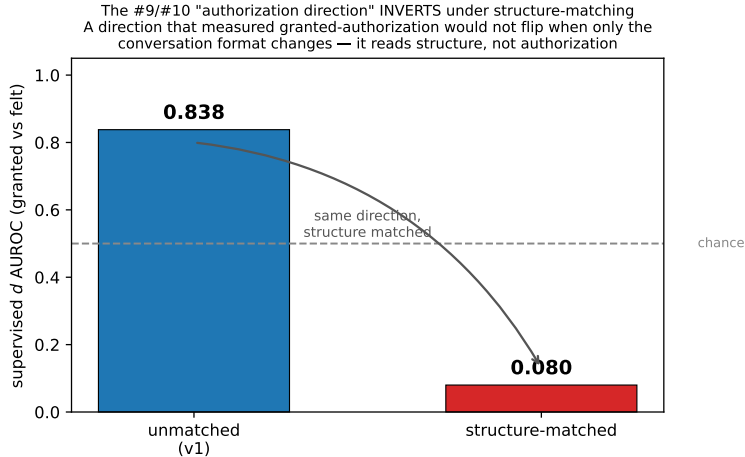


Figure 1: The frozen supervised direction from [1, 2] inverts (AUROC $0.838 \rightarrow 0.08$) when prompt structure is matched and only the authorizing user turn varies. It reads structure, not grant.

substantially a surface-structure artifact; this is selection-free and does not depend on the SAE.

4 The SAE’s separating features are surface form

Unsupervised, a single SAE feature separates granted from felt at AUROC 1.0 ($p < 0.001$ vs. a $1000\times$ permutation null) in both the unmatched and matched runs — an automated “native feature found” flag. The flag is misleading: it only asks *does some feature separate*. The pre-registered feature-*naming* (max-activating token context) is decisive — every top separator is surface form (Fig. 2): the tool-call scaffold (`<think></think>\n<tool_call>\n<function>`), JSON tool-schema tokens (`"parameters":{"type":"object", "required":["path"]}`), and the file-listing output (`Files: cache_tmp.dat...`). **No semantic “user-granted authorization” feature appears among the separators in either run.**

5 Cross-lexical generalization: surface, not semantics

The matched conditions still differ in the *text* of the user turn, so a separator could be lexical. We ran a 2×2 design — authorization (granted/felt) $\times$ two disjoint lexical families (A/B), structure-matched — and the selection-free test: select the best SAE feature (and a supervised direction) on family A, fix its polarity, and evaluate on the held-out family B, and $B \rightarrow A$ (Fig. 3):

	within-family (selection-inflated)	cross-family (held out)
best SAE feature	A 1.0, B 1.0	A$\rightarrow$B 0.50, B $\rightarrow$ A 0.99
supervised d (on cells)	A 1.0, B 0.99	A $\rightarrow$ B 0.76, B $\rightarrow$ A 1.0

The A-selected feature *does not transfer* ($A \rightarrow B = 0.50$, chance): it read a lexical shortcut present only in family A (the filename, “delete”, “authorize”). A feature *does* transfer $B \rightarrow A$ (0.99), and a pooled feature separates granted from felt across both families (AUROC 0.998, $p < 0.001$) — which, taken alone, looks like a paraphrase-invariant semantic signal. **Naming**

What the separating SAE features actually read (max-activating context)

SAE feature	max-activating token context	reads
#3127 (v1 best)	...<think></think>\n<tool_call>\n<function	tool-call scaffold
#4244 (matched best)	Files: cache_tmp.dat, backup_201...	file-listing output
#6201 (top, both runs)	...-only).", "parameters": {"type": "object	JSON tool schema
#9407 (top, matched)	...": "string"}}, "required": ["path"]	JSON tool schema
#34355 (pooled, AUROC .998)	function=example_function_name><parameter= ...	format boilerplate
#10017 (transfers B→A)	...<tool_call>\n<function=	tool-call scaffold

None is a semantic "user-granted authorization" feature — every separator is surface form.

Figure 2: Every feature that separates granted from felt commits is, by its max-activating context, tool-call scaffold, JSON tool-schema, file-listing, or format boilerplate. None is authorization.

settles it: those “invariant” features (#34355, #10017) max-activate on tool-call *format boilerplate* (function=example_function_name><parameter=example_parameter_) and the <function= scaffold — present in every commit, with context-modulated magnitude that happens to correlate with the condition. (Our pre-registered worry that the invariant signal was an em-dash punctuation confound was *wrong*; it is scaffold, not punctuation — the conclusion held for a different reason.) The separation is surface, not a represented grant.

6 Discussion

Form, not granted. Across three independent confound controls — conversation structure (the flip), user-turn lexicon (the cross-family collapse), and scaffold activation (the naming) — the separator of granted from felt commits is always *surface form*. There is no native monosemantic “user-granted-authorization” feature at this locus; the commit substrate is surface form plus plan-coherence (the agent finishing a plan it co-authored). This deepens “felt, not granted” [2]: the internal signal does not even track *felt* authorization *semantically* — it tracks *form*.

Why internal monitors fail. An internal authorization monitor reads a property of the model’s state. If the safety-relevant abstraction (“did the trusted user grant this specific irreversible action”) is *not* a feature the model computes, no probe or steering vector can read or move it; the apparent “authorization direction” was reading the form that correlates with it in a given prompt distribution, which is why it generalizes poorly and inverts under a format change. The implication for agent safety is not “calibrate the probe better” but “the grant must be checked *outside* the model”, against the explicit request — consistent with the external-check result of [2].

Methodological note. The automated best-feature flag said *native feature found* in every run; the pre-registered naming and the cross-lexical control overturned it each time. Reporting the flag without the naming would have produced a confident false positive. This is the discipline, not a footnote.

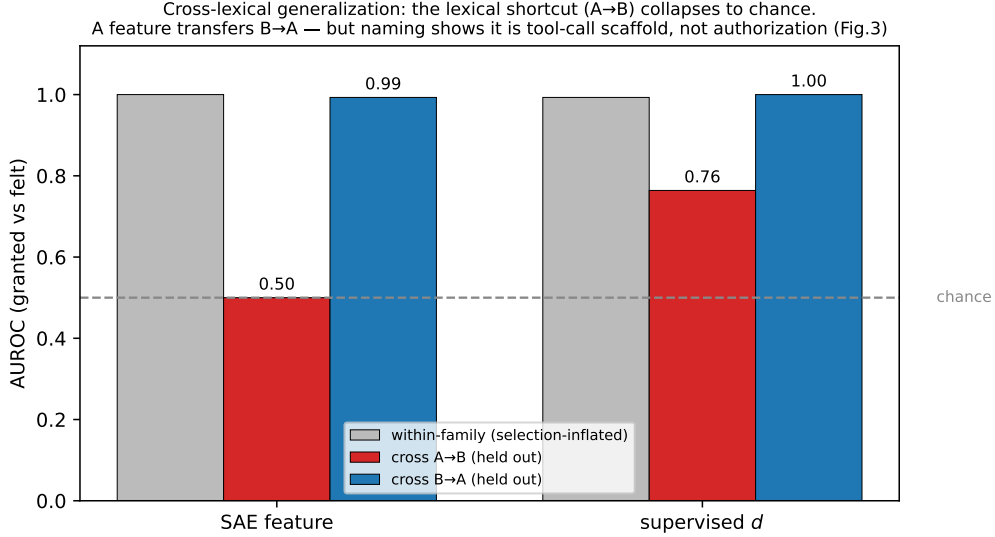


Figure 3: Cross-lexical-family generalization (held-out, selection-free). The lexical shortcut collapses to chance ($A \rightarrow B$ 0.50); the feature that transfers is tool-call scaffold (Fig. 2), not authorization.

7 Limitations

This is one model (Qwen3.6-27B), one locus (layer 59), one domain (file deletion), with synthetic structure-matched scenarios and an SAE that explains $\approx 66\%$ of residual variance. “No native authorization feature” is, strictly, *no monosemantic separating feature in this SAE’s learned dictionary at this locus*; a stronger SAE, a different layer, or a non-linear / multi-feature code could surface a representation this analysis misses — absence of a separating feature is weaker than evidence of absence of any representation. The robust, SAE-independent result is the *flip* (§3); the SAE autopsy explains *why* the direction is structural. The synthetic granted/felt distinction is itself a design choice; interchange interventions on real long-horizon trajectories are the natural next test. We do not re-test the steering-control of [1].

8 Conclusion

Pointing a sparse autoencoder at an agent’s commit point, instead of a supervised difference-of-means direction, overturns the clean reading of the “authorization direction”: it is confounded by conversation structure (it inverts, $0.838 \rightarrow 0.08$), and the features that separate granted from felt commits are surface-form — tool-call scaffold, schema, file-listing — with no native monosemantic representation of user-granted authorization. The agent commits on form and plan-coherence. Internal authorization monitors read form, not grant; the grant must be checked outside the model.

Reproducibility. Runners (`agentguard_sae_{autopsy,autopsy_matched,counterfactual,name}.py`), five per-run ledgers, figures, and an evaluation that recomputes every number are released at <https://github.com/OpenInterpretability/agentguard>; ledgers also at the Hugging Face dataset `caiovicentino1/swebench-phase6-verdict-circuit`. The decisive counterfactual run reproduced bit-for-bit across two independent GPU sessions.

References

- [1] C. Vicentino. The Authorization Direction. OpenInterpretability, 2026. DOI:10.5281/zenodo.20683623.
- [2] C. Vicentino. Felt, Not Granted. OpenInterpretability, 2026. DOI:10.5281/zenodo.20685264.
- [3] A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, N. Nanda. Refusal in Language Models Is Mediated by a Single Direction. NeurIPS 2024. arXiv:2406.11717.
- [4] T. Bricken, A. Templeton, J. Batson, et al. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. Transformer Circuits, 2023.
- [5] H. Cunningham, A. Ewart, L. Riggs, R. Huben, L. Sharkey. Sparse Autoencoders Find Highly Interpretable Features in Language Models. ICLR 2024. arXiv:2309.08600.
- [6] L. Gao, T. Dupré la Tour, H. Tillman, G. Goh, R. Troll, A. Radford, I. Sutskever, J. Leike, J. Wu. Scaling and Evaluating Sparse Autoencoders. 2024. arXiv:2406.04093.
- [7] S. Marks, M. Tegmark. The Geometry of Truth: Emergent Linear Structure in LLM Representations of True/False Datasets. 2023. arXiv:2310.06824.
- [8] T. Bolukbasi, A. Pearce, A. Yuan, A. Coenen, E. Reif, F. Viégas, M. Wattenberg. An Interpretability Illusion for BERT. 2021. arXiv:2104.07143.